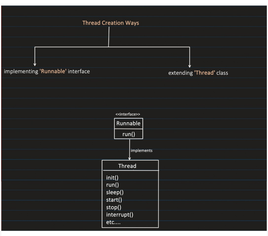
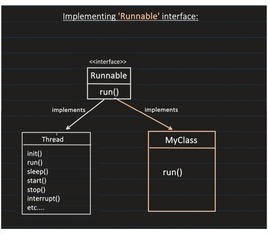
Multithreading : Part2

"Concept && Coding" YT Video Notes



Report Abuse



Step1: Create a Runnable Object

- Create a class that implements 'Runnable' interface.
- Implement the "run()" method to tell the task which thread has to do.

```
public class MultiThreadingLearning implements Runnable{

    @Override
    public void run() {
        System.out.println("code executed by thread: " + Thread.currentThread().getName());
    }
}
```

Step2: Start the thread

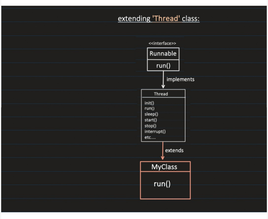
- Create an instance of class that implement 'Runnable'.
- Pass the Runnable object to the Thread Constructor.
- Start the thread.

```
public class Main {
    public static void main(String args){

        System.out.println("Going inside main method: " + Thread.currentThread().getName());
        MultiThreadingLearning runnableObj = new MultiThreadingLearning();
        Thread myThread = new Thread(runnableObj);
        myThread.start();
        System.out.println("Finish main method: " + Thread.currentThread().getName());
    }
}
```

Output:

```
Going inside main method: main
Finish main method: main
code executed by thread: Thread-0
```



Step1: Create a Thread Subclass

- Create a class that extends 'Thread' class.
- Override the "run()" method to tell the task which thread has to do.

```
public class MultiThreadingLearning extends Thread{

    @Override
    public void run() {
        System.out.println("code executed by thread: " + Thread.currentThread().getName());
    }
}
```

Step2: Initiate and Start the thread

- Create an instance of the subclass.
- Call the start() method to begin the execution.

```
public class Main {
    public static void main(String args){

        System.out.println("Going inside main method: " + Thread.currentThread().getName());
        MultiThreadingLearning myThread = new MultiThreadingLearning();
        myThread.start();
        System.out.println("Finish main method: " + Thread.currentThread().getName());
    }
}
```

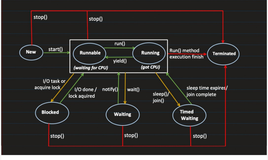
Output:

```
Going inside main method: main
Finish main method: main
code executed by thread: Thread-0
```

Why we have 2 ways to create threads?

- A class can implement more than 1 interface
  but
- A class can extend only 1 class.

Thread Lifecycle



Lifecycle State	Description
New	• Thread has been created but not started • Its just an Object in memory
Runnable	• Thread is ready to run. • Waiting for CPU time.
Running	• When thread start executing its code.
Blocked	Different scenarios where runnable thread goes into the Blocking state: - I/O : like reading from a file or database. - Lock acquired: If thread want to lock on a resource which is locked by other thread, it has to wait. • Releases all the MONITOR LOCKS
Waiting	• Thread goes into this state when we call the wait() method, makes it non runnable. • Its goes back to runnable, once we call notify() or notifyAll() method. • Releases all the MONITOR LOCKS
Timed Waiting	• Thread waits for specific period of time and comes back to runnable state, after specific conditions met. like sleep(), join(). • Do not Releases any MONITOR LOCKS
Terminated	• Life of thread is completed, it can not be started back again.

Before seeing an example, lets first understand "MONITOR LOCK"

MONITOR LOCK:
It helps to make sure that only 1 thread goes inside the particular section of code (a synchronized block or method)

```
public class MyactivityDemo{

    public synchronized void test1() {
        try {
            Thread.sleep(2);
            System.out.println("inside test1");
            Thread.dump(100L-1000L);
        } catch (Throwable e) {
            //exception handling here
        }
    }

    public void test2() {
        System.out.println("test2, Not within synchronized");
        synchronized (this) {
            System.out.println("test2, Inside synchronized");
        }
    }

    public void test3() {
        System.out.println("test3");
    }
}
```

```
public class MyactivityDemoMain {

    public static void main(String[] args) {

        MyActivityDemo obj = new MyActivityDemo();

        Thread t1 = new Thread(() -> obj.test1());
        Thread t2 = new Thread(() -> obj.test2());
        Thread t3 = new Thread(() -> obj.test3());

        t1.start();
        t2.start();
        t3.start();
    }
}
```

How this can an Example

```
public class MonitorLockExample {

    private int counter = 0;

    public void increment() {
        counter++;
        System.out.println("Incrementing counter to: " + counter);
    }

    public void decrement() {
        counter--;
        System.out.println("Decrementing counter to: " + counter);
    }

    public int getCounter() {
        return counter;
    }

    public static void main(String[] args) {
        MonitorLockExample obj = new MonitorLockExample();

        Thread t1 = new Thread(() -> {
            for (int i = 0; i < 5; i++) {
                obj.increment();
                try {
                    Thread.sleep(100);
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
            }
        });

        Thread t2 = new Thread(() -> {
            for (int i = 0; i < 5; i++) {
                obj.decrement();
                try {
                    Thread.sleep(100);
                } catch (InterruptedException e) {
                    e.printStackTrace();
                }
            }
        });

        t1.start();
        t2.start();

        try {
            t1.join();
            t2.join();
        } catch (InterruptedException e) {
            e.printStackTrace();
        }

        System.out.println("Final counter value: " + obj.getCounter());
    }
}
```

```
public class MonitorLockExample {

    private int counter = 0;

    public void increment() {
        counter++;
        System.out.println("Incrementing counter to: " + counter);
    }

    public void decrement() {
        counter--;
        System.out.println("Decrementing counter to: " + counter);
    }

    public int getCounter() {
        return counter;
    }

    public static void main(String[] args) {
        MonitorLockExample obj = new MonitorLockExample();

        Thread t1 = new Thread(() -> {
            for (int i = 0; i < 5; i++) {
                synchronized (obj) {
                    obj.increment();
                    try {
                        Thread.sleep(100);
                    } catch (InterruptedException e) {
                        e.printStackTrace();
                    }
                }
            }
        });

        Thread t2 = new Thread(() -> {
            for (int i = 0; i < 5; i++) {
                synchronized (obj) {
                    obj.decrement();
                    try {
                        Thread.sleep(100);
                    } catch (InterruptedException e) {
                        e.printStackTrace();
                    }
                }
            }
        });

        t1.start();
        t2.start();

        try {
            t1.join();
            t2.join();
        } catch (InterruptedException e) {
            e.printStackTrace();
        }

        System.out.println("Final counter value: " + obj.getCounter());
    }
}
```

```
public class Main {

    public static void main(String[] args) {

        System.out.println("Main thread start");

        Runnable runnable = new Runnable() {
            @Override
            public void run() {
                System.out.println("Runnable thread start");
                System.out.synchronized (this) {
                    System.out.synchronized (this) {
                        System.out.synchronized (this) {
                            System.out.synchronized (this) {
                                System.out.synchronized (this) {
                                    System.out.synchronized (this) {
                                        System.out.synchronized (this) {
                                            System.out.synchronized (this) {
                                                System.out.synchronized (this) {
                                                    System.out.synchronized (this) {
                                                        System.out.synchronized (this) {
                                                            System.out.synchronized (this) {
                                                                System.out.synchronized (this) {
                                                                    System.out.synchronized (this) {
                                                                        System.out.synchronized (this) {
                                                                            System.out.synchronized (this) {
                                                                                System.out.synchronized (this) {
                                                                                    System.out.synchronized (this) {
                                                                                        System.out.synchronized (this) {
                                                                                            System.out.synchronized (this) {
                                                                                                System.out.synchronized (this) {
                                                                                                    System.out.synchronized (this) {
                                                                                                        System.out.synchronized (this) {
                                                                                                            System.out.synchronized (this) {
                                                                                                                System.out.synchronized (this) {
                                                                                                                    System.out.synchronized (this) {
                                                                                                                        System.out.synchronized (this) {
                                                                                                                            System.out.synchronized (this) {
                                                                                                                                System.out.synchronized (this) {
                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                System.out.synchronized (this) {
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    System.out.synchronized (this) {
                
```

Assignment: Implement PRODUCER CONSUMER Problem

Question:

Two threads, a producer and a consumer, share a common, fixed-size buffer as a queue.
The producer's job is to generate data and put it into the buffer, while the consumer's job is to consume the data from the buffer.
The problem is to make sure that the producer won't produce data if the buffer is full, and the consumer won't consume data if the buffer is empty.